INDIVIDUAL CONTRIBUTION REPORT

Student Club Event Ticketing & Budget Portal

Software Engineering – Labs 1, 2 and 3

Report detail	Information
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1. Purpose and Reporting Basis

This report consolidates the documented responsibilities and outputs associated with the Student Club Event Ticketing & Budget Portal across Software Engineering Labs 1, 2 and 3. It explains the project context, the individual's assigned work, the Jira items allocated, the architecture model, and the next implementation and validation steps.

The report distinguishes analysis, modelling and planning evidence from software implementation evidence. A Jira item marked Done records the workflow state of that work item; by itself, it does not prove that a corresponding executable feature has been implemented and tested.

2. Project Overview

The project is a centralized portal for student-club event proposals and budgets, administrative decisions, QR-ticket generation, ticket booking and access, and registration monitoring. The intended workflow links proposal approval to ticket availability and links successful payment to ticket issuance.

Actor	Primary responsibilities documented
Club Lead	Create event proposals, enter event details and proposed budget, and submit proposals for review.
Campus Admin	Review proposals and approve or reject them.
Student	Purchase tickets for approved events, then view or download tickets.

2.1 Scope and quality expectations

- Core functional scope: submit event proposals and budgets; review and approve/reject proposals; generate unique QR tickets after approval; purchase tickets after successful payment; view/download tickets; monitor registrations and availability.
- Security expectation: role-based authentication and authorization for protected operations.
- Performance expectation: at least 95% of requests should respond within 3 seconds under expected load. This is a target to test, not a measured result established by the reviewed records.

3. Individual Responsibility and Jira Allocation

The documented Lab 3 responsibility is scenario review and requirements analysis, including functional and non-functional requirements, actor responsibilities, and use-case mapping. The allocated Jira work covers authentication/access control and event proposal/budget management.

Issue	Story	Points	Sprint	Purpose
SCRUM-13	User Login	3	Sprint 1	Credential-based login.
SCRUM-14	Role-Based Access	5	Sprint 1	Restrict features and operations according to user role.
SCRUM-15	Create Event Proposal	5	Sprint 1	Capture event details and proposed budget.
SCRUM-16	Submit Event Proposal	3	Sprint 1	Submit a proposal for administrative review.

Total allocation: 16 estimated story points. The reviewed Jira records show these four work items in Done status. That status is reported as workflow evidence, not as independent proof of implementation or test completion.

4. How the Work Was Approached

1. Started from the scenario and identified the three roles—Club Lead, Campus Admin and Student—rather than treating all users as having the same permissions.
2. Made proposal inputs explicit, including event name, date, venue, description, expected participants and proposed budget.
3. Split broad requirements into separately trackable stories for login, role-based access, creating a proposal and submitting a proposal.
4. Linked the stories to the relevant Epics, assigned priorities, estimated relative effort with story points and placed the allocated stories in Sprint 1.
5. Mapped requirements to the architecture and identified dependencies that require validation, including approval before ticket generation and successful payment before ticket issuance.

5. UML Model and Architecture

The Lab 3 records state that the team compared Layered, Microservices and Client–Server architectures and selected Layered Architecture. The documented model contains five components and six interfaces. The diagrams below visualize the recorded actors, component set, interfaces and layered organization.

5.1 Use-case model

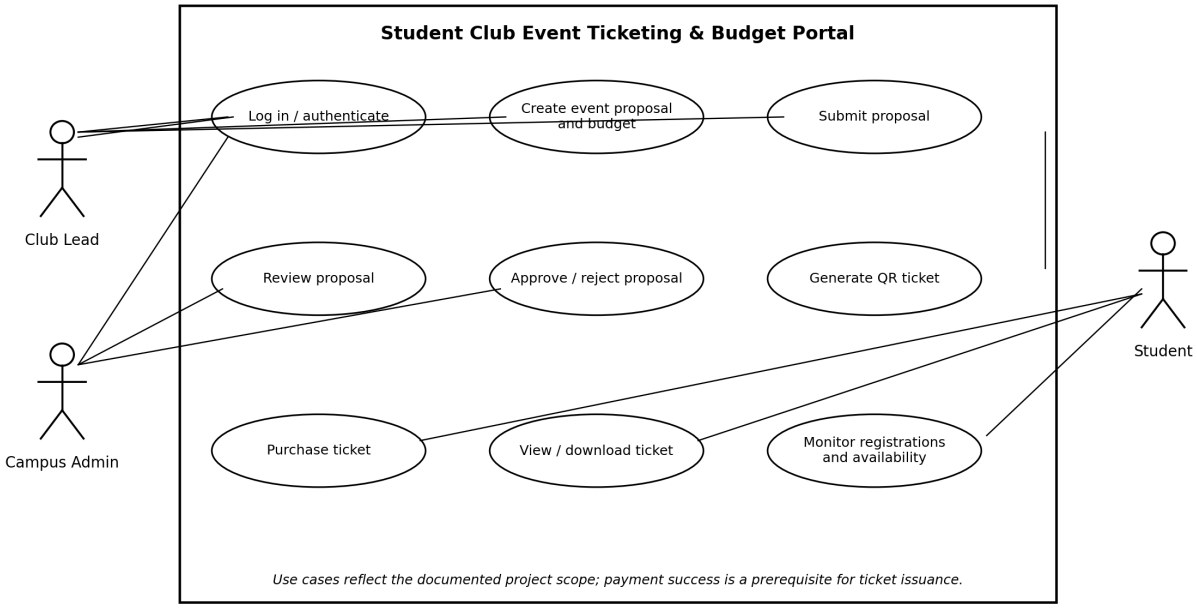


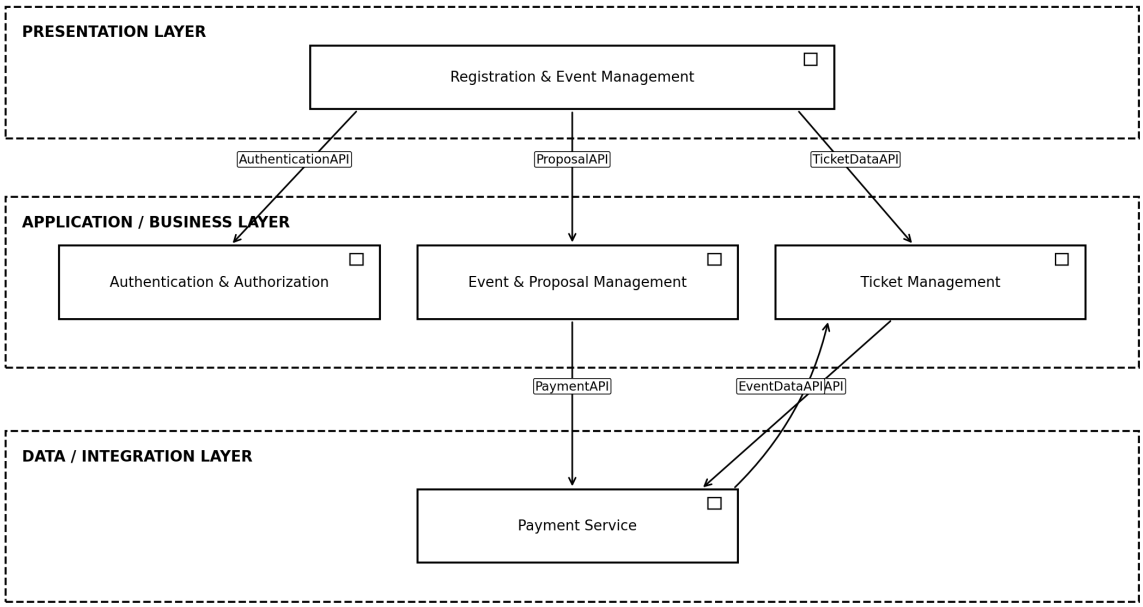
Figure 1. Use-case overview of the portal. The actors and use cases reflect the project scope documented in the report; the diagram is a high-level model, not evidence of a completed implementation.

The Club Lead is associated with authentication, proposal creation and submission. The Campus Admin is associated with proposal review and approval/rejection. The Student is associated with ticket purchase and ticket access. Registration and availability monitoring is included as a system capability. The core dependency is that an event proposal must be approved before ticket generation, and a ticket should be issued only after successful payment.

5.2 Component model

Component	Responsibility in the recorded design
Authentication & Authorization	Validate identity and enforce role-based access to protected functions.
Event & Proposal Management	Capture event details and budgets; manage proposal submission and review status.
Ticket Management	Support ticket creation/access, including QR-ticket generation after approval.
Payment Service	Coordinate or record payment outcome; ticket issuance depends on successful payment.
Registration & Event Management	Support event/registration monitoring and availability-related information.

UML Component Diagram — Layered Architecture



Interfaces shown are the six recorded contracts; arrows summarize dependency/data direction for readability.

Figure 2. UML-style component view. Six named interface contracts are shown: AuthenticationAPI, ProposalAPI, PaymentAPI, RegistrationAPI, EventDataAPI and TicketDataAPI.

The diagram groups related components by presentation, application/business, and data/integration concerns. The interface arrows are a readable summary of dependency/data direction; they are not intended to replace a fully detailed implementation-level connector specification.

5.3 Layered architecture view

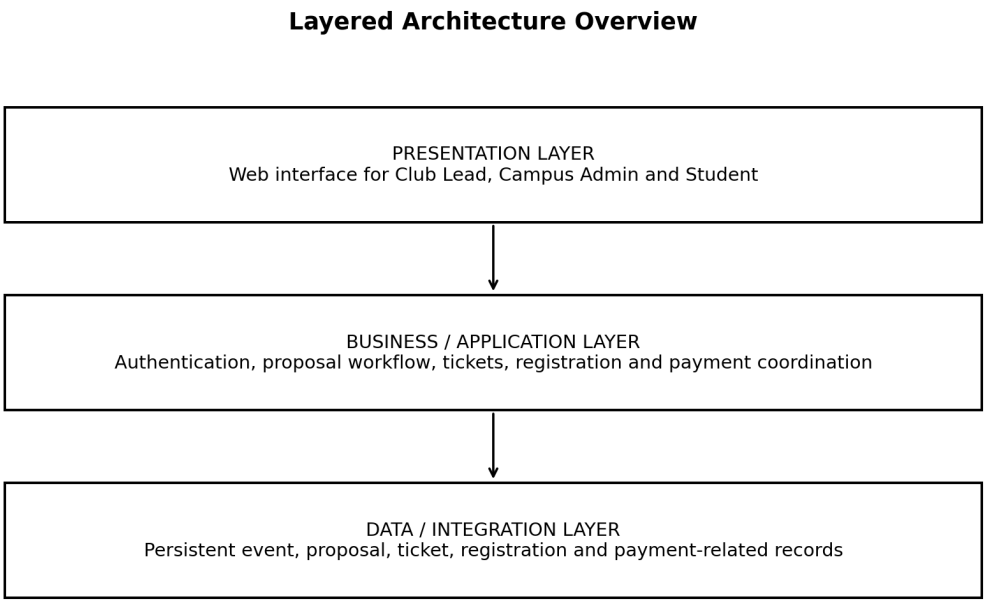


Figure 3. Layered architecture overview. The presentation layer receives user interactions, the business/application layer applies workflow rules, and the data/integration layer supports persistence and external service interactions.

5.4 Interface catalogue

Interface	Purpose / contract
AuthenticationAPI	Supports identity validation and authorization-related requests.
ProposalAPI	Supports proposal creation, submission and review workflow operations.
PaymentAPI	Supports payment requests/results used to decide whether a ticket can be issued.
RegistrationAPI	Supports registration and availability-related operations.
EventDataAPI	Provides event-related data needed by dependent components.
TicketDataAPI	Provides ticket-related data needed by dependent components.

These descriptions summarize the interface names and workflow dependencies recorded in the report. Detailed endpoint signatures, payload schemas, authentication protocols and error contracts were not included in the available source records and should be defined during implementation.

6. Lab-by-Lab Contribution Narrative

6.1 Lab 1 — Requirements Engineering and Use-case Modelling

The shared requirements foundation identifies Club Lead, Campus Admin and Student actors and covers proposal/budget submission, administrative review, ticket generation after approval, ticket purchase/access, registration

monitoring, role-based access and response-time expectations. The available records do not reliably assign exclusive authorship of Lab 1 work to an individual, so this is presented as a shared baseline.

6.2 Lab 2 — Agile Backlog Creation and Sprint Simulation

The Jira backlog contains eight Epics and sixteen user stories, with priorities, story-point estimates and team ownership. The individual allocation is SCRUM-13 through SCRUM-16, totalling 16 points. Two sprints were planned and tracked using the board, burndown, timeline and summary evidence. The allocated stories are recorded as Done in Jira.

6.3 Lab 3 — Component Modelling and Architecture Selection

The documented Lab 3 responsibility was scenario review and requirements analysis. The team compared Layered, Microservices and Client–Server architecture styles, selected Layered Architecture, identified five components and specified six interfaces. The model maps user and workflow requirements to component responsibilities and highlights approval, payment, ticketing and registration dependencies.

7. Dependencies, Risks and Validation Plan

Dependency / risk	Validation approach
Approval before ticket generation	Test approved and rejected proposals; verify that rejected or pending events cannot issue tickets.
Successful payment before ticket issuance	Test successful, failed, cancelled and timed-out payment outcomes.
Role-based access	Attempt protected actions using each role; confirm Students cannot perform Campus Admin approval actions.
Registration and event capacity consistency	Test concurrent registrations, full events and duplicate purchase attempts.
Input validation	Test missing fields, invalid event dates, malformed budget values and incomplete proposal details.
Performance expectation	Measure response times under representative load and verify whether at least 95% of requests complete within 3 seconds.

Important dependencies include approval before ticket generation, successful payment before ticket issuance, consistency between event capacity and registrations, and authorization on protected operations. Failure paths should be tested as well as normal flows.

8. Planned Work and Next Steps

6. Implement credential validation and a secure login/session mechanism.
7. Enforce role checks on protected endpoints and test that Students cannot perform Campus Admin approval actions.
8. Implement proposal field validation and a clear Submitted/Approved/Rejected status lifecycle.
9. Persist review decisions and rejection remarks, then integrate proposal data with the administrative review workflow.
10. Add positive and negative tests for invalid credentials, missing fields and unauthorized access.
11. Integrate ticket issuance with both approval state and payment outcome; then test QR-ticket access and registration consistency.

12. Collect evidence from automated tests, security tests, concurrency checks and measured performance tests before claiming the corresponding features are verified.

9. Evidence Boundary and Status

The reviewed records support the documented requirements-analysis role, Jira story allocation, architecture comparison, component model, interface list and planning activities. They do not establish that a complete production application, payment integration, QR-validation workflow or performance-tested deployment exists. Implementation and verification status should only be updated when code, test results or deployment evidence is available.

Documented / recorded	Requires separate implementation evidence
Requirements and use-case foundation	Executable authentication and proposal workflows
Jira Epics, stories, estimates and sprint planning	Integrated payment and ticket-issuance flow
Architecture comparison and component/interface model	QR validation and registration concurrency behaviour
Written reports and planned validation criteria	Automated tests, security tests and measured performance results

10. Overall Contribution Summary

Chiranth J is documented as responsible for scenario review and requirements analysis in Lab 3 and for Jira stories SCRUM-13–SCRUM-16, representing 16 estimated story points. The work connects the project's actors and requirements to an organized backlog and a layered component model. The next phase is to implement and integrate the components through defined interfaces and collect evidence that the workflows satisfy functional, security and performance expectations.

Reporting note: This report was prepared from the available team reports and Jira issue records. Where individual authorship or implementation evidence was not available, the report explicitly avoids claiming it.